

Novel Word Learning from Social Robots by Autistic and Non-Autistic Children

Novel Word Learning from Social Robots by Autistic and Non-Autistic Children

This repository contains the data, analysis code, materials, and outputs supporting the study of novel word learning from socially contingent robots in autistic and non-autistic children. The materials are provided to support transparency and computational reproducibility of the analyses reported in the manuscript.

Overview

Children learned novel object labels from a video presentation that either did or did not include a socially contingent robot. Word-learning accuracy was measured with a 12-trial forced-choice test, and engagement was coded from video across multiple behavioral channels. The autistic sample was characterized using standardized diagnostic and developmental measures (ADOS-2, MSEL, Vineland-3), which were used for sample description rather than as eligibility criteria.

Preregistration

The study was preregistered on the Open Science Framework prior to data collection: <https://osf.io/hg859/>. Departures from the preregistered protocol are documented in `preregistration/deviation_log.md`.

Repository structure

OSF_release/	
— README.md	This file
— LICENSE.md	Licensing terms for code and for
data/materials	
— CITATION.cff	Citation metadata for this deposit
— ai_use_statement.md	Generative-AI transparency note
— computational_environment.md	R and package versions
— preregistration/	Preregistration link and deviation log
— materials/	Stimuli, robot script, coding manual,

```

blank consent
├─ data/
│   └─ raw/                                Held securely, not shared (see
raw/README.md)
│   └─ deidentified/                        Shareable datasets (produced by
deidentify_release.R)
│   └─ checksums.md                        MD5 checksums for the released datasets
│   └─ data_dictionary.xlsx                Variable definitions
├─ code/                                    Analysis notebook, revision scripts,
power analysis, de-id script
└─ outputs/
    └─ figures/                            Manuscript figures
    └─ *.html                             Knitted analysis notebook

```

How to reproduce

The analysis was conducted in R 4.5.3. The primary artifact is `code/Social_Robot_Word_Learning_Analysis.Rmd`, which knits to the HTML notebook in `outputs/`. Revision analyses are reproduced by `code/revision_recompute.R` and `code/revision_model_B.R`, and the power analysis by `code/simr_power_robot_fixed.R`. Exact package versions are listed in `computational_environment.md` and in Section 12 (Session Info) of the rendered notebook.

Participant identifiers. All participant IDs in this deposit (datasets, notebook output, and figures) are re-coded public IDs that cannot be linked to the identifiers used during data collection. The re-coding is performed by `code/deidentify_release.R`; the mapping is held securely by the authors and is not deposited.

What reproduces from the shared data. The word-learning and engagement analyses (notebook Sections 5–10) run from the de-identified datasets in `data/deidentified/`. Two parts require the secure source files and therefore do not re-run from the public deposit: the interrater-reliability computation (Section 3.4, raw coding workbooks) and the covariate screening on exact age and clinical scores (Section 7.1.5; the released main dataset carries banded age and banded clinical scores instead). The full analysis trail for those sections is visible in the rendered notebook.

Data-access statement

The de-identified datasets required to reproduce the published analyses are provided in `data/deidentified/`. The raw source data are not shared because they contain individual-level clinical and developmental scores on a small autistic sample, for which the combination of quasi-identifiers carries a real re-identification risk. Requests for access to the underlying data may be directed to the corresponding author and are subject to ethics approval and a data-sharing agreement.

Ethics

The study was approved by the relevant institutional human research ethics committee. Approval reference numbers are reported in the manuscript.

Use of generative AI

See `ai_use_statement.md` for the analysis-transparency note describing the use of generative AI tools in the preparation of the code and manuscript. This note matches the declaration in the manuscript.

Citation

If you use these materials, please cite both the manuscript and the OSF deposit (see `CITATION.cff`). The OSF DOI is listed in the manuscript's Data and Code Availability statements once minted.

Contact

Corresponding author: Dr Kristyn Sommer, School of Applied Psychology, Griffith University (k.sommer@griffith.edu.au).